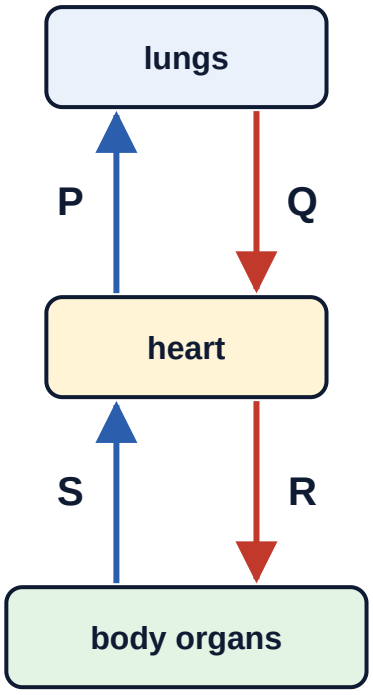
Checkpoint 6 Science — Practice Test 3 — Mark Scheme

Paper 1: 35 minutes · calculator allowed · Paper 2: 35 minutes · Total 80 marks

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 35 minutes — 40 marks

1 Hana visits the doctor's clinic for a check-up. The doctor shows her this diagram of how blood flows around the body. The arrows show which way the blood flows. **[1]**



(a) Which letter shows the blood vessel that carries blood from the heart to the body organs?

Answer: R
Notes / working: Find the box 'heart' and follow the arrow that leaves it and points **down to the body organs**: that is **R**. (S also joins the heart and the body organs, but its arrow points the other way — back to the heart.)

(b) What **type** of blood vessel is R? **[1]**

Answer: artery or arteries or an artery
Notes / working: Blood vessels that carry blood **away from the heart** are **arteries**. (Veins carry blood back to the heart.)

(c) Inside the body organs, blood flows through tiny blood vessels with walls only one cell thick. Name these blood vessels. **[1]**

> **Answer:** capillaries or capillary or capillarys
> **Notes / working:** These are **capillaries**. Their very thin walls let oxygen and nutrients pass out of the blood to the cells, and waste pass in.

(d) The nurse counts Hana's pulse for **30 seconds**. She counts 38 beats. What is Hana's pulse rate in beats per minute? **[1]**

> **Answer:** 76 beats per minute
> **Notes / working:** 1 minute = 60 seconds = 2 × 30 seconds. So beats per minute = 38 × 2 = **76**. Common mistake: giving 38 (that is only half a minute).

(e) Blood carries oxygen around the body. Name **two** other things that blood transports. **[2]**

> **Answer:** 1 mark: nutrients / digested food / glucose / sugar. 1 mark: waste / a named waste such as carbon dioxide or urea. Two wastes (e.g. carbon dioxide and urea) score 1 mark only. Do not accept: oxygen (given in the question); air; 'blood cells'.
> **Notes / working:** The circulatory system transports three things: **oxygen**, **nutrients** (digested food, e.g. glucose) and **waste** (e.g. carbon dioxide, taken to the lungs to be breathed out). So the two other things are nutrients and waste.

---

2 The doctor sees three patients. **[1]**

Patient	Illness	What caused it
Malak	athlete's foot (itchy skin between the toes)	a fungus
Karim	throat infection with a fever	bacteria
Seif	malaria	a parasite carried by mosquitoes

(a) The next patient has **flu**. What causes flu?

- ☐ A) a fungus
- ☐ B) a virus
- ☐ C) a parasite
- ☐ D) bacteria

> **Answer:** B) a virus
> **Notes / working:** Flu (influenza) and the common cold are caused by **viruses**. Diseases can be caused by viruses, bacteria, parasites or fungi, and they pass from one host to another.

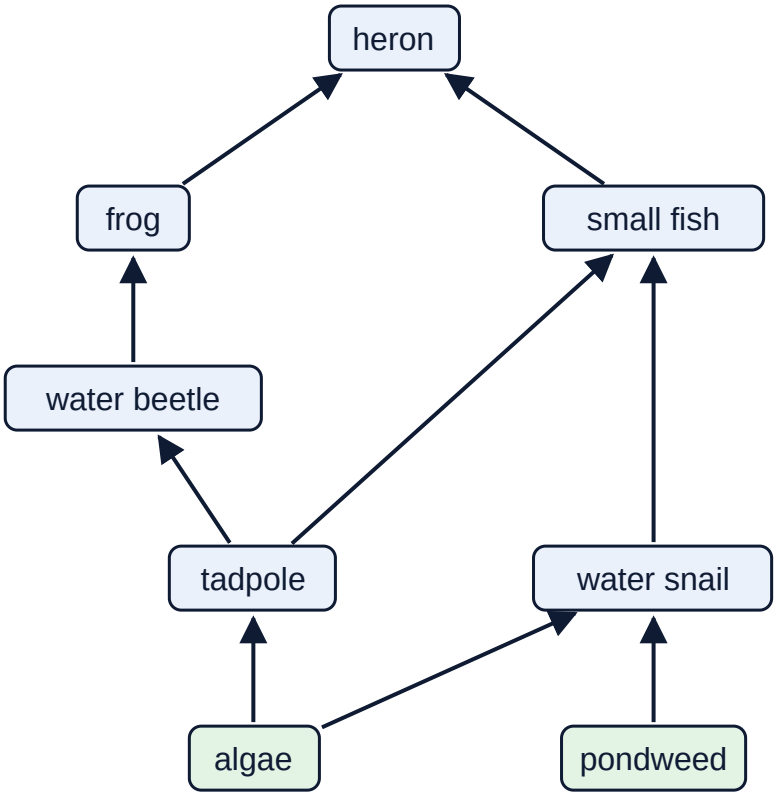
(b) Karim breathed in the bacteria. Explain how **mucus** in his nose and airways helps to protect his lungs. **[1]**

> **Answer:** 1 mark: mucus is sticky and traps pathogens / bacteria / germs / dust (so they do not reach the lungs). Accept 'catches germs'. Do not accept 'mucus kills all germs' on its own.
> **Notes / working:** Mucus is a **sticky** liquid. It **traps** bacteria, viruses and dust in the air we breathe in, so fewer pathogens reach the lungs. The mucus is then swallowed or blown out.

(c) Seif got malaria from a mosquito bite. Give **two** ways a person can avoid being bitten by mosquitoes. **[2]**

> **Answer:** 1 mark each, any two: sleep under a mosquito net; use insect repellent (spray/cream); wear long sleeves and long trousers; put screens/nets on windows or keep windows closed at night; remove or cover standing water where mosquitoes breed. Do not accept: 'wash your hands'; 'take medicine' (does not stop bites).
> **Notes / working:** Mosquitoes are **vectors**: they carry the parasite from one person to another. To stop bites: sleep under a **mosquito net**, use **insect repellent**, wear **long sleeves**, and get rid of **still water** where mosquitoes lay eggs.

3 A class studies a pond. They draw this food web. **[1]**



(a) How many food chains are there in this food web?

Answer: 4
Notes / working: Start at each producer and follow every path to the top:
1. algae → tadpole → water beetle → frog → heron
2. algae → tadpole → small fish → heron
3. algae → water snail → small fish → heron
4. pondweed → water snail → small fish → heron

So there are **4** food chains. Tip: count from **every** producer, and tick each path as you go.

(b) Name **one** producer in this food web. **[1]**

Answer: algae or alga or pondweed or pond weed or pond-weed
Notes / working: Producers make their own food using light, so they are at the start of the chains: **algae** or **pondweed**.

(c) What is the source of energy for this food web? **[1]**

Answer: sun or the sun or sunlight or light or light from the sun or the sun's light or suns light or solar energy or energy from the sun
Notes / working: The **Sun**. Plants and algae use light energy from the Sun to make food; the energy then passes along the chains when animals eat.

(d) Write down the food chain from this food web that has **five** living things. **[1]**

Answer: 1 mark: algae → tadpole → water beetle → frog → heron (all five in this order, arrows pointing towards the eater). Accept the words 'eaten by' instead of arrows. Do not accept arrows pointing the wrong way.
Notes / working: Only one chain has five links: **algae → tadpole → water beetle → frog → heron**. The arrow means 'is eaten by', so it points from the food to the animal that eats it.

(e) A poison from a nearby field washes into the pond. The algae take in the poison. It is not broken down in living things. Which animal will end up with the **most** poison in its body? **[1]**

- ☐ A) tadpole
- ☐ B) water snail
- ☐ C) water beetle
- ☐ D) heron

Answer: D) heron
Notes / working: The **heron** (the top predator). Poison passes along every food chain from the algae, and it builds up at each step.

(f) Explain your answer to part (e). **[1]**

Answer: 1 mark: the heron eats many fish / frogs (or many animals), each containing poison, so the poison builds up / adds up in its body. Accept 'it is at the top of the food chain and eats lots of animals that contain poison'. Do not accept 'it is the biggest' on its own.
Notes / working: Each small animal eats lots of algae, so it collects all their poison. The heron then eats **many** fish and frogs, so it collects the poison from **all** of them. The poison is not broken down, so it **builds up** in the top predator.

4 In the school laboratory, Leila investigates how the temperature of water affects the time a piece of chocolate takes to melt. She puts a 10 g square of chocolate in a small metal dish and floats the dish in a water bath. She times how long the chocolate takes to melt completely. **[1]**

Temperature of water in °C	Time to melt in seconds
40	240
50	150
60	190
70	70
80	50

(a) What is the **independent** variable in Leila's investigation?

Answer: temperature or temperature of the water or temperature of water or the temperature of the water or water temperature or the temperature or the water temperature or temperature of the water bath
Notes / working: The independent variable is the one Leila **changes**: the **temperature of the water**. (The time to melt is the dependent variable — the one she measures.)

(b) Give **one** variable Leila must keep the same to make this a fair test. **[1]**

Answer: 1 mark, any one: mass/size of chocolate (10 g); type/brand of chocolate; shape of the chocolate square; the same (type of) dish; amount of water in the bath. Do not accept: temperature; time; 'the same equipment' without saying which.
Notes / working: Everything except the water temperature should stay the same, for example the **mass of chocolate** (10 g each time), the **type of chocolate** or the **same metal dish**.

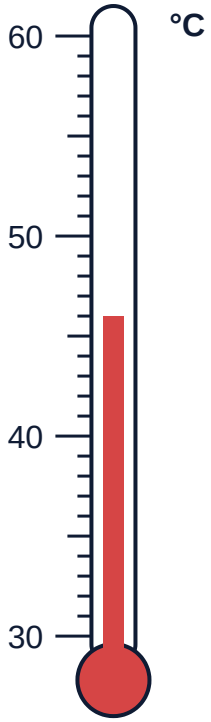
(c) One result does not fit the pattern (it is anomalous). At which temperature is the anomalous result? **[1]**

Answer: 60 °C
Notes / working: The times go down as the temperature goes up: 240, 150, … 70, 50. At **60 °C** the time (190 s) is **longer** than at 50 °C, which does not fit. We would expect a time between 150 s and 70 s.

(d) Describe the pattern in Leila's results. **[1]**

Answer: 1 mark: the higher the temperature (of the water), the shorter the time to melt / the faster the chocolate melts. Accept the reverse argument (lower temperature, longer time). Do not accept 'the time changes' (no direction).
Notes / working: The hotter the water, the less time the chocolate takes to melt. A good pattern sentence links both variables and says which way they change.

[1]



(e) Before one test, Leila reads the thermometer in the water bath. What temperature does it show?

Answer: 46 °C
Notes / working: The numbered marks are every 10 °C and there are 10 small spaces between them, so each small mark = **1 °C**. The red liquid stops 6 marks above 40, so the reading is **46 °C**.

(f) Melting chocolate is a **reversible** change. Describe how Leila could show this. **[1]**

Answer: 1 mark: cool the melted chocolate (e.g. leave it to cool / put it in a fridge) AND it becomes solid again. Do not accept 'heat it more'.
Notes / working: Let the melted chocolate **cool down** (for example in a fridge). It turns back into a **solid**, so the change can be reversed. No new substance was made.

(g) Leila puts a piece of candle wax in water at 40 °C. The chocolate melted at this temperature, but the wax does not melt. Which statement is correct? **[1]**

- ☐ A) Wax has a lower melting point than chocolate.
- ☐ B) Wax and chocolate have the same melting point.
- ☐ C) Wax has a higher melting point than chocolate.
- ☐ D) Wax can never melt.

Answer: C) Wax has a higher melting point than chocolate.
Notes / working: The temperature at which a substance melts is a **property** of that substance. Chocolate melts below 40 °C but wax does not, so **wax has a higher melting point** than chocolate. (Wax does melt — near a candle flame — so D is wrong.)

5 Ziad visits a science museum. A large model shows the Sun and the planets of the Solar System. **[1]**

(a) Which planet is the **fourth** planet from the Sun?

- ☐ A) Mars
- ☐ B) Venus
- ☐ C) Jupiter
- ☐ D) Earth

Answer: A) Mars
Notes / working: In order from the Sun: Mercury, Venus, Earth, **Mars**, Jupiter, Saturn, Uranus, Neptune. Mars is fourth.

(b) Which sentence about movement in the Solar System is correct? **[1]**

- ☐ A) The Sun orbits the Earth once a year.
- ☐ B) The Moon orbits the Sun once a month.
- ☐ C) The Earth orbits the Sun about once every 365 days.
- ☐ D) The planets orbit the Moon.

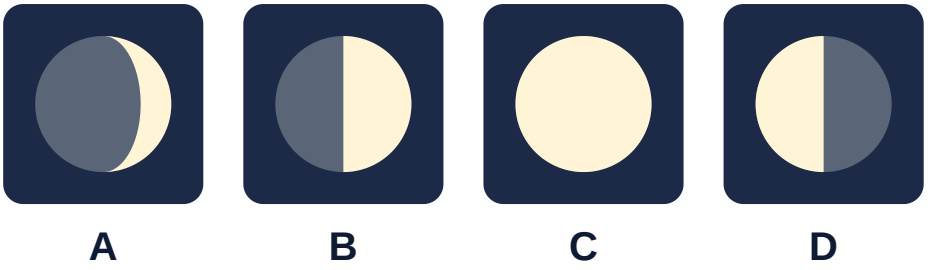
Answer: C) The Earth orbits the Sun about once every 365 days.
Notes / working: The **Earth orbits the Sun** once a year (about 365 days). The Sun is at the centre, so it does not orbit the Earth; the Moon orbits the **Earth** about once a month.

(c) What does the Moon orbit? **[1]**

Answer: earth or the earth or planet earth or the planet earth
Notes / working: The Moon orbits the **Earth**. It takes about a month to go round once.

6 (continued) **[1]**

A poster in the museum shows the Moon as seen from Egypt on four different nights.



(d) On day 0 there is a new moon. Which drawing shows the Moon about **7 days** later?

- ☐ A) A
- ☐ B) B
- ☐ C) C
- ☐ D) D

Answer: B) B

Notes / working: After a new moon, the lit part grows (waxes) on the **right** side seen from Egypt: a thin crescent first (A, about day 3–4), then a **half moon lit on the right (B) after about 7 days**, then a full moon (C) after about 15 days. D (lit on the left) comes after the full moon.

---

7 Omar rides his bicycle along a flat road. The diagram shows the two horizontal forces on Omar and his bicycle. **[1]**



(a) Work out the overall (resultant) force on Omar and his bicycle.

Answer: 40 N

Notes / working: The forces act in opposite directions, so subtract: 120 N − 80 N = **40 N** (forwards).

(b) What happens to Omar's speed? **[1]**

- ☐ A) It stays the same.
- ☐ B) It increases.
- ☐ C) It decreases.
- ☐ D) He stops at once.

Answer: B) It increases.
Notes / working: The forward force is bigger, so the overall force acts **forwards** in the direction he is moving: he **speeds up**.

(c) Later, the forward force is 100 N and the backward forces add up to 100 N. Which describes Omar's movement now? **[1]**

- ☐ A) He slows down and stops.
- ☐ B) He speeds up.
- ☐ C) He moves backwards.
- ☐ D) He keeps moving at a steady speed.

Answer: D) He keeps moving at a steady speed.
Notes / working: The forces are **balanced** (overall force = 0). A moving object with balanced forces keeps moving at a **steady speed** in the same direction. Common mistake: thinking balanced forces always mean 'not moving'.

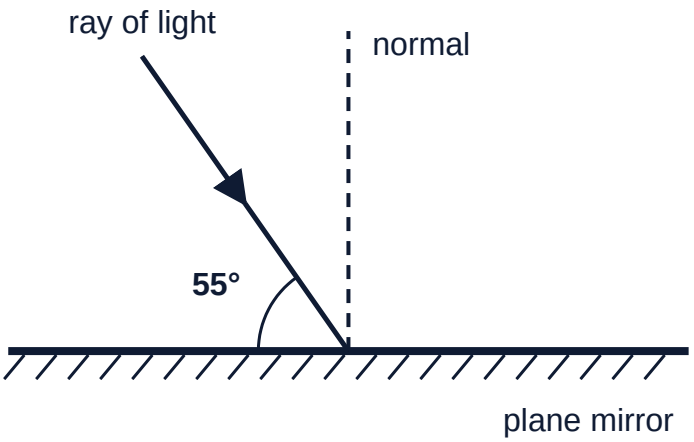
(d) Omar squeezes his brakes. Rubber brake pads press on the wheels. Name the force between the brake pads and the wheels that slows the wheels down. **[1]**

Answer: friction or friction force or frictional force or the friction
Notes / working: **Friction** acts between two surfaces that rub, and it acts against the movement.

(e) Omar stops. He and his bicycle are now still on the flat road. The weight of Omar and his bicycle is 550 N. What is the size of the upward force from the road? **[1]**

Answer: 550 N
Notes / working: An object at rest has **balanced** forces. The road pushes up with a force equal to the weight: **550 N**.

8 In the museum's light room, Ziad shines a ray of light at a plane mirror. The dashed line is the normal. **[1]**



(a) What is the angle of incidence?

Answer: 35 °
Notes / working: The angle of incidence is measured between the ray and the **normal**, not the mirror. The normal is at 90° to the mirror, so 90° − 55° = **35°**. Common mistake: writing 55°.

(b) What is the angle of reflection? **[1]**

Answer: 35 °
Notes / working: For a plane mirror, angle of reflection = angle of incidence = **35°** (ecf from part (a)).

(c) Ziad then shines a ray of light from the air into a tank of water. The ray changes direction at the surface of the water. What is this change of direction called? **[1]**

Answer: refraction or refracting or refracted or refract
Notes / working: When light passes from one medium into another (air → water) and changes direction, this is called **refraction**.

(d) Which of these is caused by refraction? **[1]**

- ☐ A) You see yourself in a bathroom mirror.
- ☐ B) A tree makes a shadow on the ground.
- ☐ C) A coin at the bottom of a glass of water looks higher than it really is.
- ☐ D) Light bounces off a shiny metal spoon.

Answer: C) A coin at the bottom of a glass of water looks higher than it really is.
Notes / working: Light from the coin bends as it passes from water into air, so the coin **looks higher**. A and D are reflection; B is a shadow (light cannot pass through the tree).

9 The museum has a rock collection. Ziad reads the information cards for three rocks. **[1]**

Rock	What it looks like
K	rounded grains stuck together in layers; some grains rub off; there is a fossil fish in it
L	large crystals of different colours that fit tightly together; no layers
M	very hard; small crystals lined up in wavy bands

(a) Which rock, K, L or M, is an **igneous** rock?

Answer: L
Notes / working: Igneous rock forms when magma cools and becomes solid, so it is made of **interlocking crystals** with no layers and no fossils: rock **L**.

(b) Which rock, K, L or M, is a **sedimentary** rock? **[1]**

Answer: K
Notes / working: Sedimentary rock is made of **grains** in **layers**, it is often crumbly and it can contain **fossils**: rock **K**. (M, with crystals in bands, is metamorphic.)

(c) Describe how the fossil fish in rock K could have formed. **[2]**

Answer: 1 mark: the fish died and (sank and) was quickly buried / covered by sediment (mud/sand) (on the bottom of a lake/sea). 1 mark: more layers built up on top and pressed / squeezed the sediment into rock OR the bones/hard parts were replaced by minerals (and turned to stone). Do not accept: 'the fish turned to rock straight away'; 'it was frozen'.
Notes / working: 1. The fish died and sank to the bottom of the water.
2. It was quickly **covered by sediment** (mud or sand), and the soft parts rotted away.
3. More layers built up on top; their weight **pressed** the lower layers into **sedimentary rock**, and **minerals** replaced the bones, making a stone copy — the fossil.

Paper 2 — 35 minutes — 40 marks

1 Nour's family goes camping in Sinai. **[1]**

(a) Nour stirs a spoonful of salt into a cup of hot water and another spoonful into a cup of cold water. The salt dissolves faster in the hot water. Which is the best explanation?

- ☐ A) The salt melts in the hot water.
- ☐ B) The water particles move faster in hot water, so the salt particles mix in among them more quickly.
- ☐ C) Hot water has fewer particles, so there is more space for the salt.
- ☐ D) The salt particles get bigger in hot water.

Answer: B) The water particles move faster in hot water, so the salt particles mix in among them more quickly.
Notes / working: In hotter water the particles have more energy and **move faster**, so they hit the salt more often and the salt particles spread out among the water particles **more quickly**. Dissolving is **not** melting (A is a common mistake).

(b) Nour hangs wet towels on a rope in the sunshine. Two hours later the towels are dry. Name the process that made the water leave the towels. **[1]**

Answer: evaporation or evaporating or evaporate or evaporated
Notes / working: The liquid water turned into water vapour from the surface of the towels without boiling: this is **evaporation**.

(c) Nour's brother says: "The water in the towels boiled away." Explain why he is wrong. **[2]**

Answer: 1 mark: boiling only happens at the boiling point / at 100 °C (for water) and the towels / air were much cooler than this. 1 mark: evaporation can happen at any temperature / below the boiling point OR evaporation happens only at the surface (boiling happens all through the liquid, with bubbles). Do not accept 'the Sun boiled it'.
Notes / working: Boiling happens only at the **boiling point** (100 °C for water) and all through the liquid, with bubbles of gas. The towels were nowhere near 100 °C. The water **evaporated**: evaporation happens **at any temperature**, only from the **surface** of a liquid.

2 The family cooks on a camping stove. Their cooking pot is made of steel. The handle of the pot is made of wood. **[1]**

(a) Why is the pot made of steel?

- ☐ A) Steel is a good thermal conductor, so heat passes quickly to the food.
- ☐ B) Steel is a thermal insulator, so it keeps the food cool.
- ☐ C) Steel is a poor conductor of electricity.
- ☐ D) Steel is a light material that floats on water.

Answer: A) Steel is a good thermal conductor, so heat passes quickly to the food.
Notes / working: Metals such as steel are **good thermal conductors**: heat moves through them quickly from the flame to the food. (Steel is actually a good electrical conductor too, so C is false.)

(b) Explain why the handle is made of wood. **[1]**

Answer: 1 mark: wood is a poor thermal conductor / a thermal insulator, so the handle stays cool / does not burn the hand. Accept 'wood does not let heat through easily'. Do not accept 'wood is strong' or 'wood is cheap'.
Notes / working: Wood is a **poor thermal conductor** (an insulator), so heat from the pot does not travel along the handle — it stays cool enough to hold.

(c) At school, Nour tests these objects in a circuit with a lamp to find out which ones are electrical conductors: **[1]**
a steel spoon, a wooden peg, a plastic cup, a strip of aluminium foil, a glass marble, a piece of copper wire.
How many of these objects will make the lamp light?

Answer: 3
Notes / working: Metals conduct electricity: the **steel** spoon, the **aluminium** foil and the **copper** wire = **3**. Wood, plastic and glass are electrical insulators.

3 The high mountains of southern Sinai are made of granite. Granite is an igneous rock. **[1]**

(a) Granite formed deep underground when magma (hot liquid rock) cooled and turned into solid rock. What is the name of this process?

Answer: solidification or solidifying or solidify or solidified or freezing or crystallisation or crystallization or cooling
Notes / working: A liquid turning into a solid is **solidification**. Magma that solidifies forms igneous rock.

(b) Wind, rain and sudden floods break the granite into small grains of sand and carry the grains away. Name this process. **[1]**

> **Answer:** erosion or eroding or erode or eroded or weathering and erosion or erosion and weathering or weathering
> **Notes / working:** Wearing rock away and carrying the pieces off is **erosion**.

(c) The sand grains settle in layers at the bottom of a lake. Over a very long time the layers are buried and pressed together into rock. Which type of rock forms? **[1]**

> **Answer:** sedimentary or sedimentary rock or a sedimentary rock
> **Notes / working:** Layers of sediment that are buried and pressed together (sedimentation and burial) form **sedimentary** rock, e.g. sandstone.

(d) Describe how this rock could change into a metamorphic rock. **[2]**

> **Answer:** 1 mark: it is buried deeper (underground) / pushed down by movements of the Earth. 1 mark: it is heated AND squeezed / high temperature and high pressure (but does not melt). Do not accept: 'it melts' (melting makes magma, which forms igneous rock) — if the answer says the rock melts, maximum 1 mark.
> **Notes / working:** The sedimentary rock is **buried deeper** underground. There it is **heated and squeezed** (high temperature and pressure) **without melting**, so its grains change into new crystals: this is **metamorphism**. If it melted, it would become magma, not metamorphic rock.

---

4 Adam visits the doctor's clinic because he has a cough. The doctor explains how his lungs work. **[1]**

(a) Which list shows the path that air takes when Adam breathes in?

- ☐ A) nose → air sacs → trachea → bronchi
- ☐ B) trachea → nose → air sacs → bronchi
- ☐ C) nose → bronchi → trachea → air sacs
- ☐ D) nose → trachea → bronchi → air sacs

> **Answer:** D) nose → trachea → bronchi → air sacs
> **Notes / working:** Air goes in through the **nose** (or mouth), down the **trachea** (windpipe), into the **bronchi** (one to each lung) and finally into the tiny **air sacs**.

(b) Complete the sentence. **[1]**
In the air sacs of the lungs, oxygen moves from the air into the _____.

> **Answer:** blood or the blood or bloodstream or the bloodstream or blood capillaries or capillaries
> **Notes / working:** Oxygen passes from the air in the air sacs into the **blood**. The blood then carries it to every part of the body.

(c) The doctor says that many vertebrates have lungs like ours. Which animal takes in oxygen with lungs? **[1]**

- ☐ A) camel
- ☐ B) shark
- ☐ C) goldfish
- ☐ D) tuna

Answer: A) camel
Notes / working: The **camel** is a mammal and breathes with lungs, like humans. Sharks, goldfish and tuna are fish: they take oxygen from water with **gills**.

(d) Adam's cough is caused by a virus. The virus spreads in tiny drops when he coughs. Give **two** things Adam can do to stop the virus spreading to his family. **[2]**

Answer: 1 mark each, any two: cover his mouth and nose (with a tissue / his elbow) when he coughs or sneezes; put used tissues in a bin; wash his hands often with soap and water; do not share cups / spoons / towels; wear a mask; keep away from others / stay at home. Do not accept: 'take antibiotics' (antibiotics do not work on viruses); 'drink water'.
Notes / working: The virus is in **body fluids** (droplets from coughs). Good hygiene stops it reaching other people: **cover coughs** with a tissue or elbow, **bin tissues**, **wash hands** with soap, and **don't share** cups or towels.

---

5 In the school laboratory, Youssef investigates what makes iron nails rust. He sets up four test tubes and leaves them for 7 days. **[1]**

Tube	What is in the tube	Rust after 7 days
A	iron nail, air and tap water	some rust
B	iron nail in boiled water with a layer of oil on top, so no air can get in	no rust
C	iron nail in dry air (a drying powder removes the water from the air)	no rust
D	iron nail, air and salty water	a lot of rust

Before he started, Youssef predicted: "The nail in tube B will rust the most, because it is completely covered in water."

(a) Which **two** tubes show that water is needed for rusting?

- ☐ A) A and B
- ☐ B) B and C
- ☐ C) A and C
- ☐ D) C and D

Answer: C) A and C
Notes / working: Compare two tubes where **only water** is different. A and C both have air; A has water and rusts, C has no water and does not rust. So **A and C**. (A and B show that **air** is needed; C and D differ in more than one way.)

(b) Rusting is a chemical reaction. Iron is one reactant. Name **one** other reactant. **[1]**

> **Answer:** oxygen or water or oxygen gas or air
> **Notes / working:** iron + oxygen + water → rust. The reactants are iron, **oxygen** (from the air) and **water**; the product is rust.

(c) What evidence shows that a chemical reaction happened in tube A? **[1]**

> **Answer:** 1 mark: a change in colour — the (shiny/grey) nail turned orange / brown / red-brown. Accept 'orange-brown rust appeared on the nail'. Do not accept 'the nail got bigger' or 'bubbles'.
> **Notes / working:** The evidence is a **change in colour**: the shiny grey nail became covered in orange-brown rust, a new substance.

(d) Was Youssef's prediction correct? Use the results to explain your answer. **[1]**

> **Answer:** 1 mark: No, AND the nail in B did not rust at all / D rusted the most, because B had no air (oxygen). 'No' alone = 0 marks.
> **Notes / working: No.** The nail in tube B did **not rust at all**, because the oil stopped air (oxygen) getting to it. The most rust was in tube D. Rusting needs **both** water and oxygen.

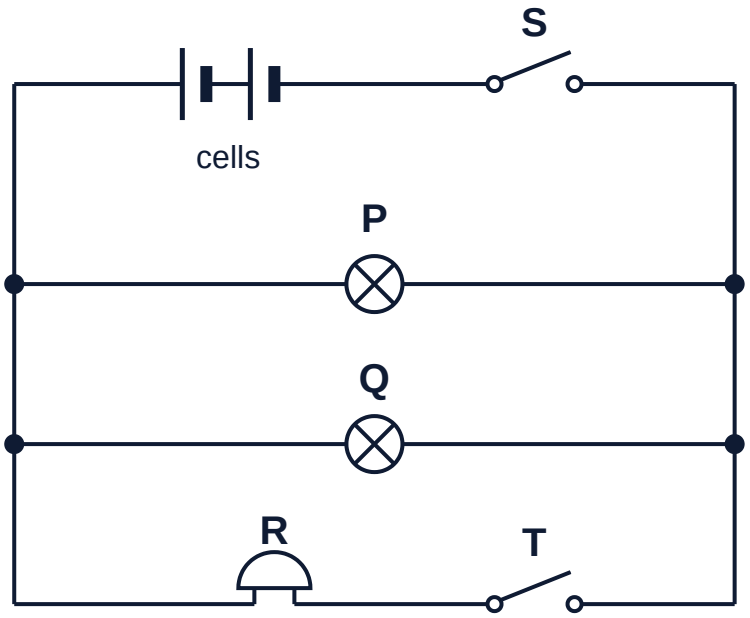
(e) Give **one** variable Youssef should keep the same in all four tubes. **[1]**

> **Answer:** 1 mark, any one: type / size / mass of nail; the time left (7 days); the temperature / where the tubes are kept; the volume of liquid; the size of the test tubes. Do not accept: water; air; salt (these are what he changes).
> **Notes / working:** Control variables include the **same type and size of nail**, the **same time** (7 days) and the **same place/temperature**.

(f) Youssef used only one nail in each tube. Suggest how he could make his results more reliable, and explain why. **[1]**

> **Answer:** 1 mark: repeat the experiment / use several nails in each tube AND a reason, e.g. so he can spot an anomalous result / check that the results are similar / one nail might be different (e.g. coated).
> **Notes / working:** Use **several nails in each tube** (or repeat the whole test). If the results agree, they are **more reliable**, and any odd (anomalous) nail can be spotted.

6 Rana designs a circuit for the lamps and the bell on her bicycle. The bell is a buzzer. Both switches are shown open. **[1]**



(a) Which letter labels the buzzer?

Answer: R
Notes / working: The buzzer symbol is a half-circle on two short legs: **R**. P and Q are lamps (a circle with a cross); S and T are switches.

(b) Are lamps P and Q joined in series or in parallel? **[1]**

Answer: parallel or in parallel or parallel circuit
Notes / working: P and Q are on **separate branches**, each joined to the main wires at junctions (dots): they are in **parallel**.

(c) Which switch or switches must be closed for the buzzer to sound? **[1]**

- ☐ A) S only
- ☐ B) T only
- ☐ C) S and T
- ☐ D) neither switch

Answer: C) S and T
Notes / working: S is in the main wire next to the cells, so it controls everything. T is only in the buzzer's branch. For the buzzer to sound, the path through R must be complete: **both S and T** must be closed.

(d) Switch S is closed. Lamp P breaks. What happens to lamp Q? **[1]**

- ☐ A) Lamp Q goes out.
- ☐ B) Lamp Q stays on.
- ☐ C) Lamp Q flashes on and off.
- ☐ D) Lamp Q gets much dimmer.

Answer: B) Lamp Q stays on.
Notes / working: In a parallel circuit each branch has its own complete path to the cells, so lamp Q **stays on**. (In a series circuit, one broken lamp would break the whole circuit.)

(e) Rana's old bicycle had the same cells, but its two identical lamps were joined in series. How does the brightness of the lamps in her new parallel circuit compare? **[1]**

- ☐ A) The lamps in the parallel circuit are brighter.
- ☐ B) The lamps are the same brightness in both circuits.
- ☐ C) The lamps in the parallel circuit are dimmer.
- ☐ D) The lamps in the parallel circuit do not light.

Answer: A) The lamps in the parallel circuit are brighter.
Notes / working: Two lamps in **series** share the push of the cells, so each is dim. In **parallel**, each lamp gets the full push of the cells, so each lamp is **brighter**.

---

7 Rana's bicycle tyre is flat. She takes out the inner tube and puts it on a balance. Its mass is 205 g. She pumps air into the inner tube until it is hard and puts it on the balance again. Its mass is now 213 g. **[1]**

(a) What mass of air did Rana pump into the inner tube?

Answer: 8 g
Notes / working: Increase in mass = 213 g − 205 g = **8 g**. This is the mass of the extra air.

(b) What does Rana's experiment show about gases? **[1]**

Answer: 1 mark: gases (air) have mass. Accept 'air has mass / air weighs something / gases are matter'. Do not accept 'air is light' or 'air takes up space' on its own.
Notes / working: The mass went up when only air was added, so **gases have mass**.

8 A display at the science museum gives the strength of gravity in four places. **[1]**

Place	Strength of gravity in N/kg
Earth	10
Moon	1.6
Mars	3.7
Jupiter	25

Ziad has a bag of sand with a mass of 2 kg.

(a) Work out the weight of the bag of sand on Earth.

> **Answer:** 20 N
> **Notes / working:** weight = mass × gravity = 2 kg × 10 N/kg = **20 N**.

(b) What is the **mass** of the bag of sand on the Moon? **[1]**

> **Answer:** 2 kg
> **Notes / working:** Mass is the amount of matter, and it does **not** change when gravity changes: still **2 kg**. Common mistake: working out 2 × 1.6 = 3.2 (that would be the weight in newtons, not the mass).

(c) Work out the weight of the bag of sand on Mars. **[1]**

> **Answer:** 7.4 N
> **Notes / working:** weight = 2 kg × 3.7 N/kg = **7.4 N**. The weight is less than on Earth because gravity on Mars is weaker.

9 At a pond, Mazen puts an empty plastic bottle with its lid on into the water. It floats. He adds sand to the bottle, 50 g at a time, and puts the lid back on each time. **[1]**

Mass of sand in g	Floats or sinks?
0	floats
50	floats
100	floats
150	floats
200	sinks

(a) Explain why the bottle sank when it had 200 g of sand in it.

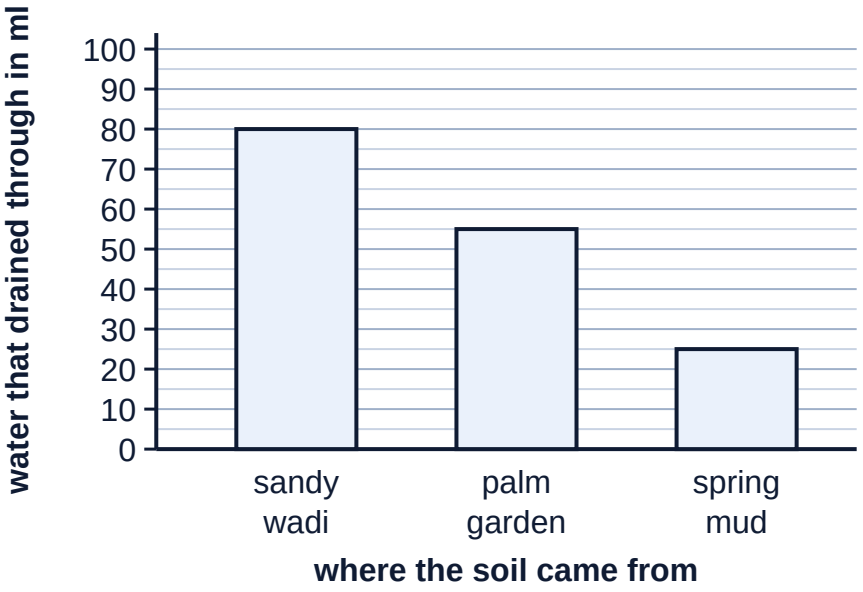
Answer: 1 mark: its weight became bigger than the upthrust (from the water). Accept 'the bottle became too heavy for its size / for the amount of water it pushes aside'. Do not accept 'it was heavy' on its own.
Notes / working: Adding sand increased the **mass** and so the **weight** of the bottle. The size (and so the upthrust) stayed the same. With 200 g of sand, the **weight was bigger than the upthrust**, so it sank.

(b) Mazen tips the 200 g of sand into a wide, empty plastic tub. The tub has the same mass as the bottle. The tub floats with the sand in it. Why? **[1]**

- ☐ A) The sand weighs less in the tub.
- ☐ B) The sand has dissolved in the water.
- ☐ C) The tub has less mass than the bottle.
- ☐ D) The tub has a different shape, so it pushes aside more water and gets more upthrust.

Answer: D) The tub has a different shape, so it pushes aside more water and gets more upthrust.
Notes / working: The total mass is the same as before, so only the **shape** has changed. The wide tub pushes aside **more water**, so the upthrust is bigger and it floats. Mass **and** shape both affect floating.

10 On the camping trip, Nour collects soil from three places: a sandy wadi (dry valley), a palm garden, and muddy ground by a spring. Back at school she puts 200 g of each soil in a funnel lined with filter paper. She pours 100 ml of water onto each soil and measures how much water drains through in 5 minutes. The bar chart shows her results. **[1]**



(a) How much water drained through the soil from the palm garden?

Answer: 55 ml
Notes / working: Each small line on the scale is 5 ml. The palm garden bar stops halfway between 50 and 60: **55 ml**.

(b) How much water did the spring mud **hold** (keep)? **[1]**

Answer: 75 ml
Notes / working: 25 ml drained through, so the soil kept 100 ml − 25 ml = **75 ml**. Common mistake: giving 25 ml (that is the water that passed through).

(c) Which soil most likely has the **most clay** in it? **[1]**

- ☐ A) the spring mud
- ☐ B) the sandy wadi soil
- ☐ C) the palm garden soil
- ☐ D) they all have the same amount of clay

Answer: A) the spring mud
Notes / working: Clay particles are very small and pack closely, so clay soil holds water and drains slowly. The **spring mud** let the least water through, so it most likely has the most clay.

(d) A farmer wants to grow vegetables in the sandy wadi soil. Suggest one change she could make to the soil to help the plants grow, and explain why it helps. **[1]**

Answer: 1 mark: add compost / manure / dead leaves (organic matter) [accept add clay] AND a reason: so the soil holds more water / gives more nutrients (minerals) to the plants. Do not accept 'add more sand' or 'add water' alone.
Notes / working: Sandy soil drains too fast and has few nutrients. Mixing in **compost** (organic matter) helps the soil **hold more water** and adds **nutrients**, so the soil composition changes in a way that **supports** plant growth.

(e) Why did Nour draw a bar chart and not a line graph? **[1]**

- ☐ A) Bar charts are always more accurate than line graphs.
- ☐ B) She only had three results, and a line graph needs ten.
- ☐ C) The amount of water is not a number.
- ☐ D) The type of soil is a word (a category), not a number that is measured.

Answer: D) The type of soil is a word (a category), not a number that is measured.
Notes / working: Use a **bar chart** when the independent variable is in **categories** (words such as types of soil). Use a **line graph** when both variables are **numbers that are measured**, e.g. temperature and time.